Problem solving with non-linear relationships

Task A: Problem solving with geometric sequences

1) Passing information is another thing which can form a geometric sequence. The bus is going to be late back from a school trip. One pupil phones their parent. The parent phones 5 other parents. Each of those parents phone 5 other parents. Assume it takes 5 minutes to make a phone call.

Time	People
10 mins	5
20 mins	25
30 mins	125
40 mins	625 + 1

×5
×5
×5

a) How many people would know after 40 mins?

626 people would know if the growth continued at the above rate.

b) Why might this not be realistic?

Some parents might ring the same people. Some parents might not ring anyone.

It might take some parents longer than 10 minutes to ring 5 parents.

Time	People
10 mins	5
20 mins	25
30 mins	125
40 mins	625
50 mins	3125
60 mins	15 625
70 mins	78 125

Time	People
10 mins	5
20 mins	5^2
30 mins	5^3
40 mins	5^4
50 mins	5^5
60 mins	5^6
120 mins	5^{12}

c) How many people would know after 2 hours?

≈ 200 000 000 people

d) Why might this not be realistic?

There will not be that many parents to tell.

Task B: Adding sequences

1) In each question, the sequences were created by adding the terms of two of the sequences given below. Work out which two sequences were added in each question.

A 1, 2, 4, 8, 16 …	B 4, 12, 36, 108, …	C 4, 9, 14, 19, …	D 1, 3, 5, 7, 9 …
E 4, 7, 12, 19, …	F 1, 6, 16, 31, 51, …	G 4, 6, 6, 4, 0 …	H 1, 8, 27, 64, …

a) 5, 12, 19, 26, 33, … *C + D*

b) 5, 11, 18, 27, 40, … *A + C*

c) 5, 13, 28, 50, 79, … *E + F*

d) 2, 9, 21, 38, 60, … *D + F*

e) 2, 5, 9, 15, 25, … *A + D*

f) 2, 11, 32, 71, 134, … *D + H*

g) 5, 14, 40, 127, 356, … *A + B*

h) 8, 15, 20, 23, 24, … *C + G*

i) 8, 13, 18, 23, 28, … *E + G*

2) What is the n^{th} term rule for sequence a). How does that relate to the two sequences you have chosen for your answer to a)?

$7n - 2$. *The* n^{th} *term rules for C and D are* $5n - 1$ *and* $2n - 1$.

Adding the n^{th} *terms together:* $5n - 1 + 2n - 1 = 7n - 2$.

3) By using the examples previously and any of your own examples, investigate the following questions and write a sentence about any patterns you have noticed.

a) What happens when you add two geometric sequences together?

E.g. You do not get another geometric sequence. There is no clear arithmetic or geometric pattern in the differences or ratios between terms.

b) What happens when you add two quadratic sequences together?
(square and triangle numbers are examples of quadratic sequences)

E.g. Often you get another quadratic sequence with the common second difference of the new sequence equal to the sum of the second differences of the original sequences. If the second differences of two sequences are the additive inverses of each other it is possible to add two quadratic sequences to get a linear sequence- Question 1i) is an example of this.

c) What happens when you add a linear sequence to a geometric sequence?

E.g. The resulting sequences are neither arithmetic or geometric. There seems to be geometric relationships in the second differences between terms.

d) What happens when you add a linear sequence to a quadratic sequence?

E.g. You will get a quadratic sequence. If you knew the n^{th} *term rules, adding them together would give you the new* n^{th} *term rule.*